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Course –ECA5601- 5g/6g Communication

EXPERIMENT 4: OFDMA Resource Block Allocation

AIM: To simulate and analyze OFDMA (Orthogonal Frequency Division Multiple Access) resource block allocation and evaluate system performance in terms of resource utilization and user throughput using MATLAB.

Objective

1. To determine the Number of Resource Blocks (RBs) allocated in an OFDMA system based on available bandwidth and system configuration.
2. To evaluate the Resource Block Utilization (%) by analyzing how efficiently available RBs are assigned to active users
3. To calculate the User Throughput (Mbps) based on allocated resource blocks and modulation efficiency in the OFDMA system.

Procedure

1. Define the system bandwidth and calculate the number of available resource blocks (RBs) based on RB size (e.g., 180 kHz per RB).
2. Identify the number of active users in the OFDMA system for resource sharing.
3. Allocate resource blocks to each user based on a predefined or equal distribution strategy.
4. Compute Resource Block Utilization (%) using the ratio of allocated RBs to total RBs.
5. Assign modulation efficiency values (e.g., QPSK, 16-QAM, 64-QAM) for each user.
6. Calculate User Throughput (Mbps) using allocated RBs and modulation efficiency.
7. Plot graphs for RB allocation and utilization/throughput analysis using MATLAB.
8. Analyze results to evaluate spectrum efficiency and system performance.

objective 1 :

Program

```
clc; clear; close all;

% System parameters

bandwidth = 10e6;

rb_size = 180e3;

% Total Resource Blocks

% 10 MHz

% 180 kHz per RB

total_RB = floor(bandwidth / rb_size);

% Users

users = 1:6;

% RB allocation (simple equal + remainder distribution)

base = floor(total_RB / length(users));

alloc = base * ones(1,length(users));

alloc(1:mod(total_RB,length(users))) = alloc(1:mod(total_RB,length(users))) + 1;

utilization = (alloc / total_RB) * 100;

disp('Total Resource Blocks:')

disp(total_RB)

disp('RB Allocation per User:')

disp(alloc)

figure

% ----- Graph 1: RB Allocation per User -----

subplot(2,1,1)

stem(users, alloc, 'filled', 'LineWidth', 2)

title('OFDMA Resource Block Allocation per User')

xlabel('User Index')

ylabel('Number of Resource Blocks')

grid on
```

% ----- Graph 2: RB Utilization -----

subplot(2,1,2)

plot(users, utilization, '-o', 'LineWidth', 2)

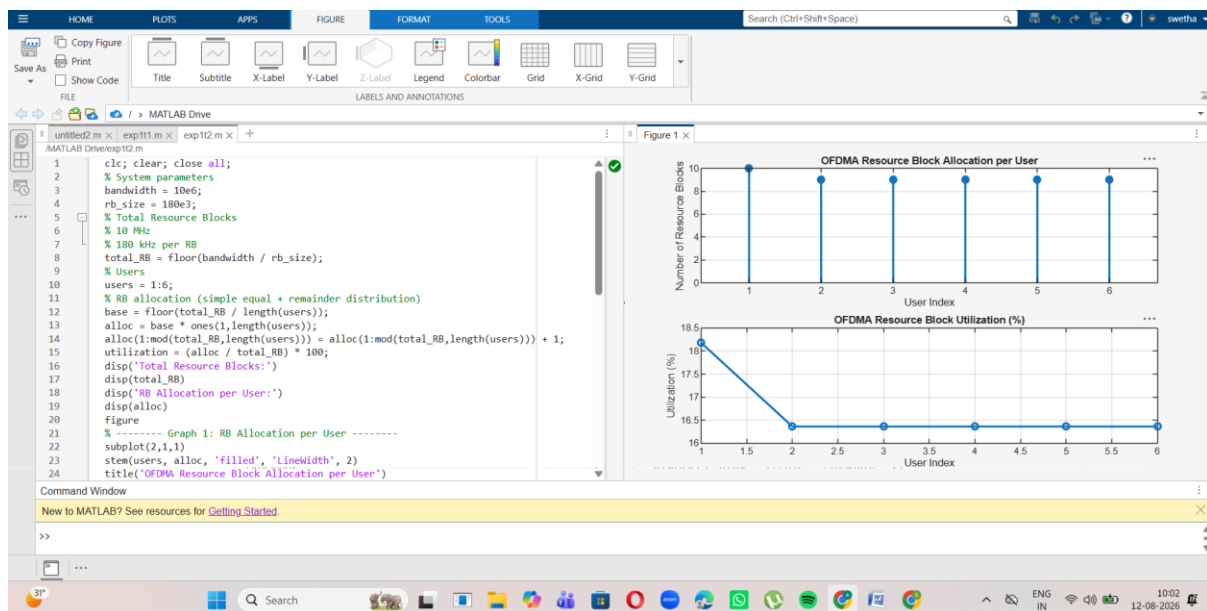
title('OFDMA Resource Block Utilization (%)')

xlabel('User Index')

ylabel('Utilization (%)')

grid on

output



Objective 2: code and output:

clc; clear; close all;

% Total system resource blocks

total_RB = 50;

% Number of active users

users = 1:5;

% RB allocation to users (example scenario)

alloc = [12 10 9 8 11];

% Resource Block Utilization (%)

util = (alloc / total_RB) * 100;

disp('Resource Block Utilization (%) per User:')

disp(util)

figure

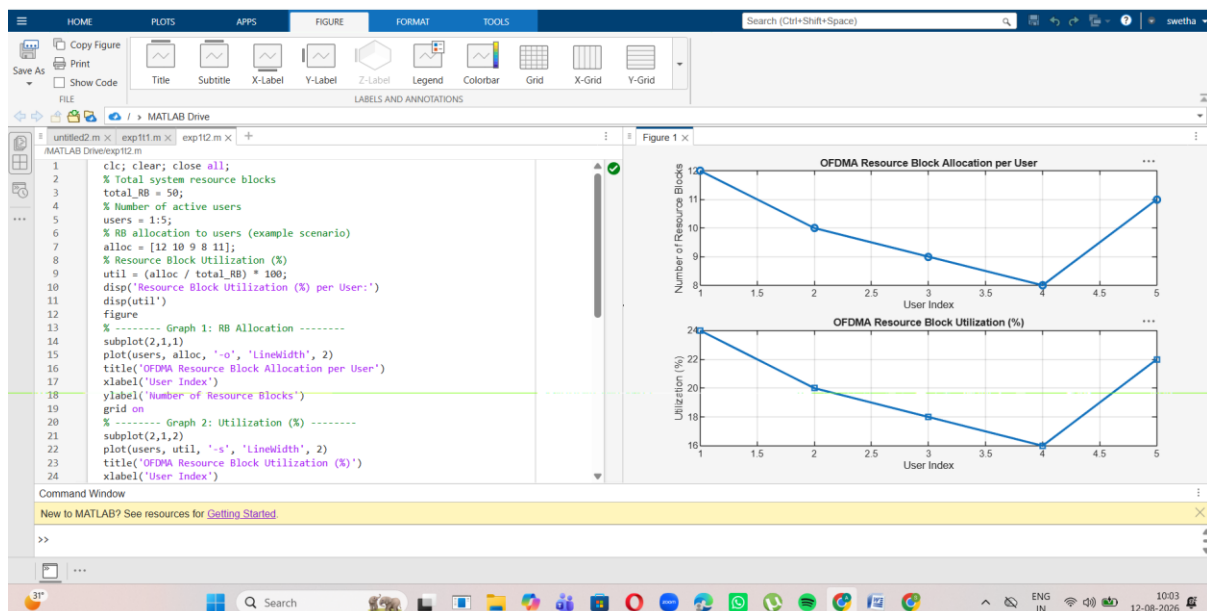
% ----- Graph 1: RB Allocation -----

subplot(2,1,1)

```

plot(users, alloc, '-o', 'LineWidth', 2)
title('OFDMA Resource Block Allocation per User')
xlabel('User Index')
ylabel('Number of Resource Blocks')
grid on
% ----- Graph 2: Utilization (%) -----
subplot(2,1,2)
plot(users, util, '-s', 'LineWidth', 2)
title('OFDMA Resource Block Utilization (%)')
xlabel('User Index')
ylabel('Utilization (%)')
grid on

```



Objective 3: code and output:

```

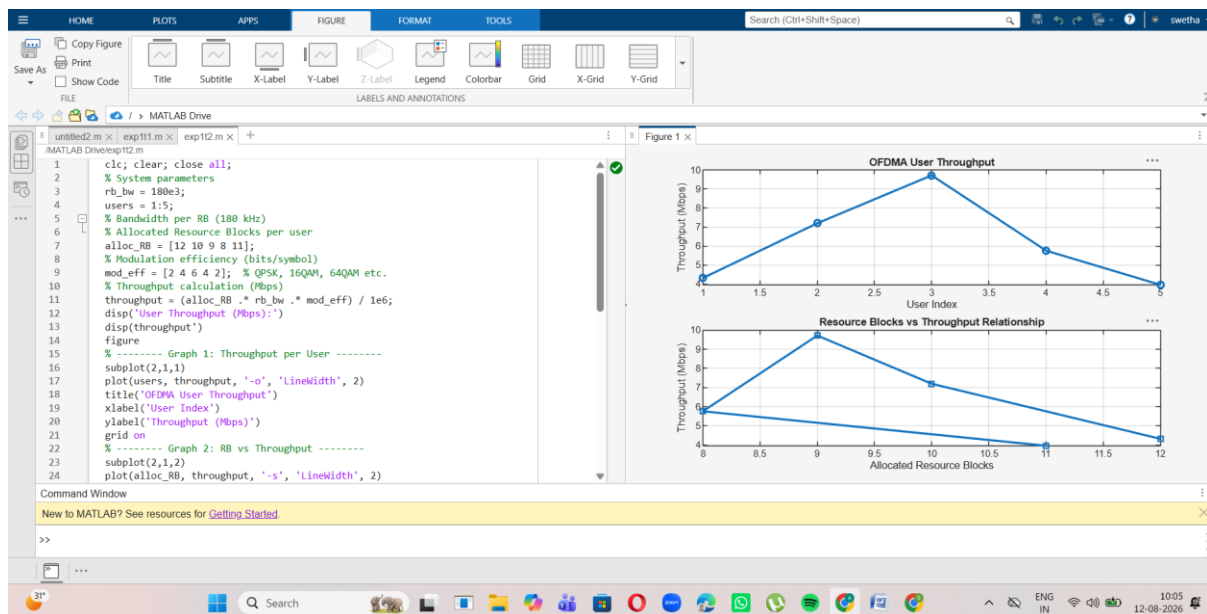
clc; clear; close all;
% System parameters
rb_bw = 180e3;
users = 1:5;
% Bandwidth per RB (180 kHz)
% Allocated Resource Blocks per user
alloc_RB = [12 10 9 8 11];
% Modulation efficiency (bits/symbol)
mod_eff = [2 4 6 4 2]; % QPSK, 16QAM, 64QAM etc.
% Throughput calculation (Mbps)
throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6;

```

```

disp('User Throughput (Mbps):')
disp(throughput')
figure
% ----- Graph 1: Throughput per User -----
subplot(2,1,1)
plot(users, throughput, '-o', 'LineWidth', 2)
title('OFDMA User Throughput')
xlabel('User Index')
ylabel('Throughput (Mbps)')
grid on
% ----- Graph 2: RB vs Throughput -----
subplot(2,1,2)
plot(alloc_RB, throughput, '-s', 'LineWidth', 2)
title('Resource Blocks vs Throughput Relationship')
xlabel('Allocated Resource Blocks')
ylabel('Throughput (Mbps)')
grid on

```



Result:

1. The Resource Block (RB) allocation was successfully simulated for multiple users in the OFDMA system, demonstrating how available spectrum is distributed among active users.
2. The Resource Block Utilization (%) analysis showed efficient usage of system resources, indicating how effectively the available RBs are assigned to users.
3. The User Throughput (Mbps) was successfully calculated based on RB allocation and modulation efficiency, showing that higher allocated RBs and higher modulation schemes result in increased throughput.